

RAKESH MANDAL

PhD Scholar, Department of Applied Mechanics, Indian Institute of Technology Delhi

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RESEARCH SUMMARY

I develop generative AI models for engineering design that produce high-performing structures in seconds and stay reliable under manufacturing uncertainty. My PhD work uses flow matching, neural surrogates and robust optimisation to replace slow, solver-in-the-loop design with learned generative priors, for 2D and 3D topology optimisation and engineering design under uncertainty. Before my PhD, I worked as a production engineer at Hero MotoCorp, where I saw first-hand how manufacturing variation turns good designs into rejected parts.

Interests: generative models (flow matching, diffusion) · topology and shape optimisation · robust and reliability-based design under uncertainty · neural surrogates and operator learning · energy-efficient and multimodal generative design

EDUCATION

PhD (Direct, after B.Tech.), Applied Mechanics Jul 2025 – present (expected 2029)
Indian Institute of Technology (IIT) Delhi New Delhi

- ▶ Advisor: Prof. Souvik Chakraborty
- ▶ Thesis (working title): *Multimodal, Robust, Sustainable and Reliability-Aware Generative Design*
- ▶ Coursework CGPA 9.50/10, department rank 2: Optimization Methods; Reliability Analysis and Reliability-Based Design Optimization; Finite Element Method; Probabilistic Machine Learning for Mechanics

PhD student (Research Training Programme), Design and Manufacturing Aug 2024 – Jul 2025
Indian Institute of Science (IISc), Bengaluru Bengaluru

- ▶ Graduate coursework in solid mechanics, materials behaviour, manufacturing science and computational mathematics before moving to IIT Delhi in July 2025

B.Tech., Mechanical Engineering 2018 – 2022
National Institute of Technology (NIT) Durgapur Durgapur, West Bengal

- ▶ CGPA 8.24/10
- ▶ Final-year project: *Observing and Modelling Nonlinear Dynamics in an Internal Combustion Engine*, under Prof. Sirshendu Mondal

Class XII (WBCHSE), 2018: 83.8% · Class X (WBBSE), 2016: 84.57%, first position in school

PUBLICATIONS

*Equal contribution. †Corresponding author.

Journal articles under review

[J1] **R. Mandal**^{*†}, A. P. Padhi^{*}, A. Dasgupta, S. Chakraborty[†]. *A Conditional Flow-Matching Framework for Generative Topology Optimization in Two and Three Dimensions*. Under review, *Computer Methods in Applied Mechanics and Engineering* (CMAME).

RESEARCH EXPERIENCE

Doctoral Researcher Jul 2025 – present
Department of Applied Mechanics, IIT Delhi (advisor: Prof. Souvik Chakraborty) New Delhi

TopoFlow: generative topology optimisation in 2D and 3D [J1]

- ▶ Built a single conditional flow-matching model (U-Net velocity field with Fourier-feature and FiLM conditioning) that generates 2D and 3D optimised structures without architectural changes.
- ▶ 2D: matches classical SIMP compliance (median ratio 1.008), with 93.2% of designs within 10% of the SIMP reference, at 25–100× lower cost.
- ▶ 3D (SELTO benchmarks): volume-fraction error 1.3–4.1%; best-of-16 sampling yields a connected design within 10% of the FEM reference in up to 96.5% of test cases, about 27–54× faster than SIMP

INDUSTRY EXPERIENCE

Graduate Engineer Trainee (Assistant Manager role), Production Jul 2022 – Jul 2024
Hero MotoCorp Ltd. Halol Plant, Gujarat

- ▶ Led the team that reduced crankshaft rejections in the induction-hardening operation by 70%.

- ▶ Planned daily production and manpower; ran cost-optimisation and rejection-reduction projects using structured problem solving on the shop floor.
- ▶ Pillar head of the GreenCo (green company) initiative; deputy leader of the Development Management pillar of Total Productive Maintenance (TPM); strengthened the 5S red-tag system.

UNDERGRADUATE PROJECTS AND TRAINING

- ▶ **Nonlinear dynamics in an IC engine** (B.Tech. project, Prof. Sirshendu Mondal): observed and modelled nonlinear combustion dynamics.
- ▶ **Cellular-automaton model of particle motion and its applications** (mini project, Prof. Sirshendu Mondal).
- ▶ **SAE BAJA, Team NDORS, NIT Durgapur**: team member at SAE BAJA 2020 (Pithampur, Indore); Technical Head, Powertrain, 2021.
- ▶ **Industrial internship, MSME-CTTC Bhubaneswar** (2 weeks): turning, milling, drilling and CNC machining; built a hand-injection mould; CATIA training.

SKILLS

Machine learning	PyTorch (mixed-precision GPU training), flow matching, diffusion models, 2D/3D U-Nets, FiLM and classifier-free guidance, neural operators, surrogate modelling, Bayesian optimisation
Computational design	Topology optimisation (SIMP), finite-element analysis, robust design optimisation, uncertainty quantification, SDF/voxel geometry processing
Programming & tools	Python (NumPy, SciPy, pandas, trimesh, scikit-image), MATLAB, Git, $\LaTeX$, Linux GPU workstations (NVIDIA A40), Weights & Biases
CAD	CATIA, SolidWorks, AutoCAD
Languages	English, Hindi, Bengali

LEADERSHIP AND ACHIEVEMENTS

- ▶ Department rank 2 in PhD coursework (CGPA 9.50/10), Department of Applied Mechanics, IIT Delhi, 2025–26.
- ▶ Led the induction-hardening rejection-reduction project at Hero MotoCorp (70% reduction).
- ▶ Technical Head, Powertrain, Team NDORS (SAE BAJA), NIT Durgapur, 2021.
- ▶ First position in school, West Bengal Secondary (Madhyamik) Examination, 2016.